

Ch3. Matrix algebra

3.1 Vector and matrix

■ A brief review of vector and matrix

Remark 3.1 $\underline{x}_{n \times 1} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ $\underline{A}_{n \times n} = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix}_{n \times n}$

Rank ($\underline{A}$)

* square matrix $\underline{A}_{n \times n}$ special case $\underline{A} = \underline{A}'$ symmetric

non-negative or positive $\underline{A} \geq 0$

ex. $\underline{A} = \text{cov}(\underline{x}) = \begin{pmatrix} \text{cov}(x_1, x_1) & \dots & \text{cov}(x_1, x_n) \\ \vdots & & \vdots \\ \text{cov}(x_n, x_1) & \dots & \text{cov}(x_n, x_n) \end{pmatrix} \geq 0$

* 2f Rank($\underline{A}$) = $\overset{n \times n}{n}$ (full rank)

$\Rightarrow$ unique $\underline{A}^{-1}$: $\underline{A} \underline{A}^{-1} = \underline{A}^{-1} \underline{A} = \underline{I}_{n \times n}$

$$- \underline{A} \underline{x} = \underline{c} \Rightarrow \underline{x} = \underline{A}^{-1} \underline{c}$$

$$- (\underline{A} \underline{B})^{-1} = \underline{B}^{-1} \underline{A}^{-1}$$

* Partition of a matrix: $\underline{A} = \begin{pmatrix} \underline{A}_{11} & \underline{A}_{12} \\ \underline{A}_{21} & \underline{A}_{22} \end{pmatrix}$

$$\underline{A}^{-1} = \begin{pmatrix} \underline{A}_{11}^{-1} + \underline{A}_{11}^{-1} \underline{A}_{12} \underline{B}^{-1} \underline{A}_{21} \underline{A}_{11}^{-1} & -\underline{A}_{11}^{-1} \underline{A}_{12} \underline{B}^{-1} \\ -\underline{B}^{-1} \underline{A}_{21} \underline{A}_{11}^{-1} & \underline{B}^{-1} \end{pmatrix}$$

$$\underline{B} = \underline{A}_{22} - \underline{A}_{21} \underline{A}_{11}^{-1} \underline{A}_{12}$$

$$= \begin{pmatrix} \underline{D}^{-1} & -\underline{D}^{-1} \underline{A}_{12} \underline{A}_{22}^{-1} \\ -\underline{A}_{22}^{-1} \underline{A}_{21} \underline{D}^{-1} & \underline{A}_{22}^{-1} + \underline{A}_{22}^{-1} \underline{A}_{21} \underline{D}^{-1} \underline{A}_{12} \underline{A}_{22}^{-1} \end{pmatrix}$$

$$\underline{D} = \underline{A}_{11} - \underline{A}_{12} \underline{A}_{22}^{-1} \underline{A}_{21}$$

- Ex: special cases. $\underline{A} = \begin{pmatrix} \underline{A}_{11} & \underline{a}_{12} \\ \underline{a}_{12} & a_{22} \end{pmatrix}$ $\underline{a}_{12}$ - vector

$$\underline{A}^{-1} = \frac{1}{b} \begin{pmatrix} b \underline{A}_{11}^{-1} + \underline{A}_{11}^{-1} \underline{a}_{12} \underline{a}_{12}^{-1} \underline{A}_{11}^{-1} & -\underline{A}_{11}^{-1} \underline{a}_{12} \\ -\underline{a}_{12}^{-1} \underline{a}_{11}^{-1} & 1 \end{pmatrix}$$

$$b = a_{22} - \underline{a}_{12}^{-1} \underline{A}_{11}^{-1} \underline{a}_{12}$$

- Ex. special case

$$\begin{aligned} \underline{B}^{-1} &= (\underline{A}_{22} - \underline{A}_{21} \underline{A}_{11}^{-1} \underline{A}_{12})^{-1} \\ &= \underline{A}_{22}^{-1} + \underline{A}_{22}^{-1} \underline{A}_{21} (\underline{A}_{11} - \underline{A}_{12} \underline{A}_{22}^{-1} \underline{A}_{21})^{-1} \underline{A}_{12} \underline{A}_{22}^{-1} \end{aligned}$$

$\underline{D}$

e.g.

$$(\underline{A} + \underline{c} \underline{c}')^{-1} = \underline{A}^{-1} - \frac{\underline{A}^{-1} \underline{c} \underline{c}' \underline{A}^{-1}}{1 + \underline{c}' \underline{A}^{-1} \underline{c}}$$

$n \times n \quad n \times 1$

hint

$$\begin{pmatrix} -1 & \underline{c}' \\ \underline{c} & \underline{A}_{n \times n} \end{pmatrix}$$

3.1 Vector and matrix – Full Rank Factorization

Theorem : $\underline{A}_{p \times q}$ of rank r can always be factorized as

$$\underline{A} = \underline{K}_{p \times r} \underline{L}_{r \times q}$$

where $\underline{K}$ and $\underline{L}$ have full column and full row rank respectively.

Proof: There exist nonsingular matrices $\underline{P}$ and $\underline{Q}$ such that

$$\underline{P} \stackrel{d}{=} \begin{pmatrix} \underline{U}_{r \times p} \\ \underline{V}_{(p-r) \times p} \end{pmatrix} \quad \underline{P} \underline{A} \underline{Q} = \begin{bmatrix} \underline{I}_r & \underline{0} \\ \underline{0} & \underline{0} \end{bmatrix} \quad \underline{Q} \stackrel{d}{=} \begin{pmatrix} \underline{R} & \underline{S} \\ \underline{0} & \underline{0} \end{pmatrix}$$

$$\Rightarrow \underline{A} = \underline{P}^{-1} \begin{bmatrix} \underline{I}_r & \underline{0} \\ \underline{0} & \underline{0} \end{bmatrix} \underline{Q}^{-1}.$$

Partition $\underline{P}^{-1}$ and $\underline{Q}^{-1}$ as

$$\underline{P}^{-1} \stackrel{d}{=} \begin{bmatrix} \underline{K}_{p \times r} & \underline{W}_{p \times (p-r)} \end{bmatrix}, \quad \underline{Q}^{-1} \stackrel{d}{=} \begin{bmatrix} \underline{L}_{r \times q} \\ \underline{Z}_{(q-r) \times q} \end{bmatrix}.$$

3.1 Vector and matrix – Full Rank Factorization

Then,

$$\begin{aligned} A &= [K \ W] \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} L \\ Z \end{bmatrix} \\ &= [K \ 0] \begin{bmatrix} L \\ Z \end{bmatrix} \\ &= KL. \end{aligned}$$

Note: the concept of left inverse and the right inverse.

$$\underline{P} \stackrel{d}{=} \begin{pmatrix} \underline{U} \\ \underline{V} \end{pmatrix} \quad \underline{P}^{-1} = \begin{pmatrix} \underline{K} & \underline{W} \end{pmatrix} \Rightarrow \underline{P} \underline{P}^{-1} = \begin{pmatrix} \underline{U} \underline{K} & \underline{U} \underline{W} \\ \underline{V} \underline{K} & \underline{V} \underline{W} \end{pmatrix} = \underline{I}_{p \times p}$$

$$\Rightarrow \underline{U} \underline{K} = \underline{I}_r \quad \text{then } \underline{U} \text{ is the left inverse of } \underline{K}$$

$r \times p \quad p \times r$

$$\text{Similarly, } \underline{R} \text{ is right inverse of } \underline{L} \text{ if } \underline{L} \underline{R} = \underline{I}_r$$

$r \times q \quad q \times r$

3.1 Vector and matrix – Idempotent Matrices

$$\mathbf{A}^2 = \mathbf{A}$$

- All idempotent matrices (except $\mathbf{I}$) are singular.

Proof: Since $\mathbf{A}^2 = \mathbf{A}$ and if $\mathbf{A}$ is nonsingular,

$$\mathbf{A} = \mathbf{A}^{-1} \mathbf{A} \mathbf{A} = \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}.$$

- $r(\mathbf{A}) = \text{tr}(\mathbf{A})$.

Proof: Consider the full rank factorization, let

$$\mathbf{A} = \mathbf{B} \mathbf{C} \quad \text{and} \quad \mathbf{A}^2 = \mathbf{B} \mathbf{C} \mathbf{B} \mathbf{C} = \mathbf{B} \mathbf{C}.$$

But $\mathbf{B}$ has a left inverse $\mathbf{U}$ and $\mathbf{C}$ has a right inverse $\mathbf{R}$, then

$$\begin{aligned} \mathbf{U} \mathbf{B} \mathbf{C} \mathbf{B} \mathbf{C} \mathbf{R} &= \mathbf{U} \mathbf{B} \mathbf{C} \mathbf{R} \\ \Rightarrow \mathbf{C} \mathbf{B} &= \mathbf{I}_{r \times r} \end{aligned}$$

So, $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{B} \mathbf{C}) = \text{tr}(\mathbf{C} \mathbf{B}) = \text{tr}(\mathbf{I}_{r \times r}) = r = r(\mathbf{A})$.

3.1 Vector and matrix – Idempotent Matrices

(self-study)

- Eigenvalues of idempotent matrices are either 0 or 1.

Proof: Let $\lambda, \mathbf{x}$ be a pair of eigenvalue and eigenvector.

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

$$\Rightarrow \mathbf{A}^2\mathbf{x} = \mathbf{A}(\mathbf{A}\mathbf{x}) = \mathbf{A}(\lambda\mathbf{x}) = \lambda\mathbf{A}\mathbf{x} = \lambda^2\mathbf{x}.$$

But also

$$\mathbf{A}^2\mathbf{x} = \mathbf{A}\mathbf{x} = \lambda\mathbf{x},$$

thus,

$$\lambda^2\mathbf{x} = \lambda\mathbf{x} \Rightarrow \lambda(\lambda - 1)\mathbf{x} = 0.$$

Since $\mathbf{x} \neq 0 \Rightarrow \lambda = 0$ or 1 .

3.1 Vector and matrix – Idempotent Matrices *(self-study)*

- For symmetric matrix $\mathbf{A}$, if all eigenvalues are 1 or 0, $\mathbf{A}$ is idempotent

Proof: For $\mathbf{A}$ symmetric, there exists an orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{D},$$

where $\mathbf{D}$ is a diagonal matrix with eigenvalues of $\mathbf{A}$ on the diagonal. So, $\mathbf{P}'\mathbf{A}\mathbf{P}\mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{D}^2$, But, $\mathbf{P}'\mathbf{A}\mathbf{P}\mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{P}'\mathbf{A}\mathbf{A}\mathbf{P}$. However if all eigenvalues are 1 or 0,

$$\Rightarrow \mathbf{D} = \mathbf{D}^2$$

$$\Rightarrow \mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{P}'\mathbf{A}\mathbf{A}\mathbf{P}$$

$$\Rightarrow \mathbf{A} = \mathbf{A}\mathbf{A}$$

$$\Rightarrow \mathbf{A} \text{ is idempotent}$$

3.2 Generalized Inverse

Definition: Let $\mathbf{A}$ be $m \times n$ and the generalized inverse $\mathbf{A}^-$ satisfies

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$$

$m \times n$ $n \times m$ $m \times n$

■ g-inverse may not be unique.

Example 3.1

$$\mathbf{A} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}_{4 \times 1}$$

$$\hat{\mathbf{A}}_1^- = (1, 0, 0, 0)_{1 \times 4}$$

$$\hat{\mathbf{A}}_2^- = (0, \frac{1}{2}, 0, 0)$$

$$\hat{\mathbf{A}}_3^- = (0, 0, \frac{1}{3}, 0)$$

$$\hat{\mathbf{A}}_4^- = (0, 0, 0, \frac{1}{4})$$

} are all generalized inverse of $\hat{\mathbf{A}}$.

g-inverse

3.2 Generalized Inverse

Theorem. Suppose $\mathbf{A}$ is $n \times p$ of rank r and $\mathbf{A}$ is partitioned by

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$$

where $\mathbf{A}_{11}$ is $r \times r$ of rank r (full rank). Then a generalized inverse of $\mathbf{A}$ is given by

$$\mathbf{A}^- = \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \cdot \text{is one of g-inverse of } \mathbf{A}$$

Proof.

Remark 3.2

Corollary. Suppose $\mathbf{A}$ is $n \times p$ of rank r , and $\mathbf{A}_{22}$ is $r \times r$ of rank r (full rank). Then a generalized inverse of $\mathbf{A}$ is given by

$$\mathbf{A}^- = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22}^{-1} \end{pmatrix}.$$

Remark 3.2 Need to prove $\underline{A} \underline{A}^+ \underline{A} = \underline{A}$

$$\underline{A} \underline{A}^+ \underline{A} = \begin{pmatrix} \underline{I} & \underline{O} \\ \underline{A}_{21} \underline{A}_{11}^{-1} & \underline{O} \end{pmatrix} \begin{pmatrix} \underline{A}_{11} & \underline{A}_{12} \\ \underline{A}_{21} & \underline{A}_{22} \end{pmatrix} = \begin{pmatrix} \underline{A}_{11} & \underline{A}_{12} \\ \underline{A}_{21} & \underline{A}_{21} \underline{A}_{11}^{-1} \underline{A}_{12} \end{pmatrix}$$

N.T.P. $\underline{A}_{21} \underline{A}_{11}^{-1} \underline{A}_{12} = \underline{A}_{22}$

Let $\underline{B} = \begin{pmatrix} \underline{I} & \underline{O} \\ -\underline{A}_{21} \underline{A}_{11}^{-1} & \underline{I} \end{pmatrix}$ — non-singular

$$\underline{B} \underline{A} = \begin{pmatrix} \underline{A}_{11} & \underline{A}_{12} \\ \underline{O} & \underline{A}_{22} - \underline{A}_{21} \underline{A}_{11}^{-1} \underline{A}_{12} \end{pmatrix} \quad \begin{array}{l} \text{rank}(\underline{B} \underline{A}) \\ = \text{rank}(\underline{A}) \\ = r \end{array}$$

$\text{rank}(\underline{A}_{11}) = r$

They are linear combination of the 1st r columns.

$$\Rightarrow \underline{A}_{22} - \underline{A}_{21} \underline{A}_{11}^{-1} \underline{A}_{12} = \underline{O}$$

$$\Rightarrow \underline{A} \begin{pmatrix} \underline{A}_{11}^{-1} & \underline{O} \\ \underline{O} & \underline{O} \end{pmatrix} \underline{A} = \underline{A}$$

$$\Rightarrow \begin{pmatrix} \underline{A}_{11}^{-1} & \underline{O} \\ \underline{O} & \underline{O} \end{pmatrix} \text{ is a } g\text{-inverse of } \underline{A} = \begin{pmatrix} \underline{A}_{11} & \underline{A}_{12} \\ \underline{A}_{21} & \underline{A}_{22} \end{pmatrix}$$

$$\underline{A} = \begin{pmatrix} \underline{A}_{11} & \underline{A}_{12} \\ \underline{A}_{21} & \underline{A}_{22} \end{pmatrix} \quad \text{rank}(\underline{A}) = r$$

if $\underline{A}_{22}$ is $r \times r$, $\underline{A}_{22}$ is full rank

$$\Rightarrow \begin{pmatrix} \underline{O} & \underline{O} \\ \underline{O} & \underline{A}_{22}^{-1} \end{pmatrix} \text{ is another } g\text{-inverse of } \underline{A}$$

$$\underline{A} = \begin{pmatrix} \underline{A}_{11} & \underline{A}_{12} \\ \underline{A}_{21} & \underline{A}_{22} \end{pmatrix} \quad \text{rank}(\underline{A}) = r$$

if $\underline{A}_{21}$ is $r \times r$, $\underline{A}_{21}$ is full rank.

$$\Rightarrow \begin{pmatrix} \underline{O} & \underline{A}_{21}^{-1} \\ \underline{O} & \underline{O} \end{pmatrix} \text{ is a } g\text{-inverse of } \underline{A}$$

3.2 Generalized Inverse

Example 3.2

$$A = \left(\begin{array}{cc|c} 2 & 2 & 3 \\ 1 & 0 & 1 \\ 3 & 2 & 4 \end{array} \right)$$

$$\text{rank}(A) = 2$$

$$A_{11} = \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix}$$

$$A_{22} = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}$$

$$A_{21} = \begin{pmatrix} 1 & 0 \\ 3 & 2 \end{pmatrix}$$

$$A_1^- = \begin{pmatrix} A_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{2} & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

is a g-inverse
of A

$$A_2^- = \begin{pmatrix} 0 & 0 \\ 0 & A_{22}^{-1} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -2 & \frac{1}{2} \\ 0 & 1 & 0 \end{pmatrix}$$

$$A_3^- = \begin{pmatrix} 0 & A_{21}^{-1} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & -\frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}$$

3.2 Generalized Inverse – properties

- Let $\mathbf{X}$ be $m \times n$, $r(\mathbf{X}) = k > 0$,
 - (i) $r(\mathbf{X}^-) \geq k$;
 - (ii) $\mathbf{X}^- \mathbf{X}$ and $\mathbf{X} \mathbf{X}^-$ are idempotent;
 - (iii) $r(\mathbf{X}^- \mathbf{X}) = r(\mathbf{X} \mathbf{X}^-) = k$;
 - (iv) $\mathbf{X}^- \mathbf{X} = \mathbf{I}$ if and only if $r(\mathbf{X}) = n$;
 - (v) $\mathbf{X} \mathbf{X}^- = \mathbf{I}$ if and only if $r(\mathbf{X}) = m$;
 - (vi) $\text{tr}(\mathbf{X}^- \mathbf{X}) = \text{tr}(\mathbf{X} \mathbf{X}^-) = k = r(\mathbf{X})$;
 - (vii) If $\mathbf{X}^-$ is any g-inverse of $\mathbf{X}$, then $(\mathbf{X}^-)'$ is a g-inverse of $\mathbf{X}'$.

3.2 Generalized Inverse – properties

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \varepsilon_i$$

$$\underline{y} = \underline{X} \underline{\beta} + \underline{\varepsilon}$$

$$\hat{\underline{\beta}} = (\underline{X}'\underline{X})^{-} \underline{X}'\underline{y}$$

- Let $\underline{K} = \underline{X}(\underline{X}'\underline{X})^{-}\underline{X}'$, $\underline{K}$ is invariant for any g-inverse of $\underline{X}'\underline{X}$.

Remark 3.3

- For $\underline{K} = \underline{X}(\underline{X}'\underline{X})^{-}\underline{X}'$,
 - (i) $\underline{K} = \underline{K}'$, $\underline{K} = \underline{K}^2$ (So, Symmetric Idempotent);
 - (ii) $\text{rank}(\underline{K}) = \text{rank}(\underline{X}) = r$ ($\text{rank}(\underline{K}) = \text{tr}(\underline{K}) = \text{rank}(\underline{X})$);
 - (iii) $\underline{K}\underline{X} = \underline{X}$, $\underline{X}'\underline{K} = \underline{X}'$;
 - (iv) $(\underline{X}'\underline{X})^{-}\underline{X}'$ is a g-inverse of $\underline{X}$ for any g-inverse of $\underline{X}'\underline{X}$;
 - (v) $\underline{X}(\underline{X}'\underline{X})^{-}$ is a g-inverse of $\underline{X}'$ for any g-inverse of $\underline{X}'\underline{X}$.

Remark 3.3. To prove $\underline{K} = \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'$ is invariant!

Proof. From the definition of g-inverse, we have

$$\underline{X}'\underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{X} = \underline{X}'\underline{X}$$

$$\Rightarrow \underline{L}'\underline{J}'\underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{J}\underline{L} = \underline{L}'\underline{J}'\underline{J}\underline{L}$$

Let $\underline{R}$ is a right inverse of $\underline{L}$,

$$\text{i.e. } \underline{L}\underline{R} = \underline{I}_r.$$

$$\Rightarrow \underline{R}'\underline{L}'\underline{J}'\underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{J}\underline{L}\underline{R} = \underline{R}'\underline{L}'\underline{J}'\underline{J}\underline{L}\underline{R}$$

$$\Rightarrow \underline{J}'\underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{J} = \underline{J}'\underline{J}$$

$$\Rightarrow \underline{J}'\underline{J}\underline{L}(\underline{X}'\underline{X})^{-1}\underline{L}'\underline{J}'\underline{J} = \underline{J}'\underline{J}$$

Note that $\underline{J}'\underline{J}$ is non-singular, we have

$$\underline{L}(\underline{X}'\underline{X})^{-1}\underline{L}' = (\underline{J}'\underline{J})^{-1}$$

$$\Rightarrow \underline{J}\underline{L}(\underline{X}'\underline{X})^{-1}\underline{L}'\underline{J}' = \underline{J}(\underline{J}'\underline{J})^{-1}\underline{J}'$$

$$\Rightarrow \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}' = \underline{J}(\underline{J}'\underline{J})^{-1}\underline{J}'$$

$\underline{J}(\underline{J}'\underline{J})^{-1}\underline{J}'$ is invariant, because

$$\begin{aligned}\underline{J}(\underline{J}'\underline{J})^{-1}\underline{J}' &= \underline{J}^* \underline{\Sigma} (\underline{\Sigma}' \underline{J}^{*'} \underline{J}^* \underline{\Sigma})^{-1} \underline{\Sigma}' \underline{J}^{*'} \\ &= \underline{J}^* (\underline{J}^{*'} \underline{J}^*) \underline{J}^{*'}\end{aligned}$$

Full rank factorization

$$\underline{X}_{m \times n} = \underline{J}_{m \times r} \underline{L}_{r \times n}$$

$$= \underline{J}_{m \times r}^* \underline{L}_{r \times n}^*$$

$$r = \text{rank}(\underline{X})$$

There exists a non-singular

$$\underline{\Sigma}_{r \times r}, \text{ s.t.}$$

$$\underline{J}_{m \times r} = \underline{J}_{m \times r}^* \cdot \underline{\Sigma}_{r \times r}$$

✗

3.3 Moore–Penrose Inverse

Definition: Let $\mathbf{A}$ be an $m \times n$ matrix. If a matrix $\mathbf{A}^+$ exists and satisfies

- $$\left. \begin{array}{l} (1) \mathbf{A}\mathbf{A}^+ \text{ is symmetric} \\ (2) \mathbf{A}^+\mathbf{A} \text{ is symmetric} \\ (3) \mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A} \\ (4) \mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+ \end{array} \right\}$$

$\mathbf{A}^+$ is defined as a Moore–Penrose inverse of $\mathbf{A}$.

Theorem Each matrix $\mathbf{A}$ has an $\mathbf{A}^+$.

Proof: If $\mathbf{A} = \mathbf{0}$ then $\mathbf{A}^+ = \mathbf{0}$.

If $\mathbf{A} \neq \mathbf{0}$, $\mathbf{A}$ can be factored by full rank factorization

$\mathbf{A} = \mathbf{A}_L \mathbf{A}_R = \mathbf{B}\mathbf{C}$, where $\mathbf{B}$ is $m \times r$ of rank r and $\mathbf{C}$ is $r \times n$ of rank r . Hence, $\mathbf{B}'\mathbf{B}$ and $\mathbf{C}\mathbf{C}'$ are both nonsingular.

Define

$$\mathbf{A}^+ = \mathbf{C}'(\mathbf{C}\mathbf{C}')^{-1}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'$$

and it can be shown that $\mathbf{A}^+$ satisfies the 4 conditions above.

3.3 Moore–Penrose Inverse: properties

- The Moore–Penrose inverse is unique;
- $(\mathbf{A}')^+ = (\mathbf{A}^+)'$;
- $(\mathbf{A}^+)^+ = \mathbf{A}$;
- $r(\mathbf{A}^+) = r(\mathbf{A})$;
- If $\mathbf{A} = \mathbf{A}'$, then $\mathbf{A}^+ = (\mathbf{A}^+)'$;
- If $\mathbf{A}$ is nonsingular, $\mathbf{A}^{-1} = \mathbf{A}^+$;
- If $\mathbf{A}$ is symmetric idempotent, $\mathbf{A}^+ = \mathbf{A}$;
- If $r(\mathbf{A}_{m \times n}) = m$, then $\mathbf{A}^+ = \mathbf{A}'(\mathbf{A}\mathbf{A}')^{-1}$, $\mathbf{A}\mathbf{A}^+ = \mathbf{I}$,
If $r(\mathbf{A}_{m \times n}) = n$, then $\mathbf{A}^+ = (\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'$, $\mathbf{A}^+\mathbf{A} = \mathbf{I}$;
- The matrices $\mathbf{A}\mathbf{A}^+$, $\mathbf{A}^+\mathbf{A}$, $\mathbf{I} - \mathbf{A}\mathbf{A}^+$ and $\mathbf{I} - \mathbf{A}^+\mathbf{A}$ are all symmetric idempotent.

3.4 Vector and matrix calculus

3.4.1. $u = f(\underline{x})$ $\underline{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix}$

$$\frac{\partial u}{\partial \underline{x}} = \begin{pmatrix} \frac{\partial u}{\partial x_1} \\ \vdots \\ \frac{\partial u}{\partial x_p} \end{pmatrix}$$

— Let $u = \underline{a}' \underline{x} = \underline{x}' \underline{a}$, then $\frac{\partial u}{\partial \underline{x}} = \frac{\partial (\underline{a}' \underline{x})}{\partial \underline{x}} = \underline{a}$

— Let $u = \underline{x}' \underline{A} \underline{x}$ $\underline{A}$ - symmetric

then $\frac{\partial u}{\partial \underline{x}} = \frac{\partial (\underline{x}' \underline{A} \underline{x})}{\partial \underline{x}} = 2 \underline{A} \underline{x}$

3.4.2. $u = f(\underline{x})$ $\underline{x} = \begin{pmatrix} x_{11} & \dots & x_{1p} \\ \vdots & & \\ x_{p1} & \dots & x_{pp} \end{pmatrix}$ $p \times p$

$$\frac{\partial u}{\partial \underline{x}} = \begin{pmatrix} \frac{\partial u}{\partial x_{11}} & \dots & \frac{\partial u}{\partial x_{1p}} \\ \vdots & & \\ \frac{\partial u}{\partial x_{p1}} & \dots & \frac{\partial u}{\partial x_{pp}} \end{pmatrix}$$

— Let $u = \text{tr}(\underline{X} \underline{A})$

$\underline{X}_{p \times p}$ — symmetric
positive definite
 $\underline{A}_{p \times p}$ — matrix of constants

$$\frac{\partial u}{\partial \underline{x}} = \frac{\partial \text{tr}(\underline{X} \underline{A})}{\partial \underline{x}} = \underline{A} + \underline{A}' - \text{diag}(\underline{A})$$

Note that $p \times p$
 $\text{tr}(\underline{X} \underline{A}) = \sum_{i=1}^p \sum_{j=1}^p x_{ij} \cdot a_{ji}$
and $x_{ij} = x_{ji}$ if $i \neq j$

$$— \frac{\partial \log |\underline{X}|}{\partial \underline{x}} = 2 \underline{X}^{-1} - \text{diag}(\underline{X}^{-1})$$

consider the case
of $p=2$

3.4.3. $\underline{A}_{n \times n} = (a_{ij})$

a_{ij} — function of x_{ik}
 $\underline{A}$ — positive definite

$$— \frac{\partial \underline{A}^{-1}}{\partial \underline{x}} = - \underline{A}^{-1} \frac{\partial \underline{A}}{\partial \underline{x}} \underline{A}^{-1}$$

$$\underline{\underline{A}} \underline{\underline{A}} = \underline{I}$$

$$\frac{\partial \underline{\underline{A}}^{-1}}{\partial \underline{x}} \underline{\underline{A}} + \underline{\underline{A}}^{-1} \frac{\partial \underline{\underline{A}}}{\partial \underline{x}} = \underline{0}$$

$$— \frac{\partial \log |\underline{A}|}{\partial \underline{x}} = \text{tr}(\underline{A}^{-1} \frac{\partial \underline{A}}{\partial \underline{x}})$$